
Lab 3 - Written Justification

Self-Service Coffee Kiosk System

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Architecture Selection

We chose Layered Architecture for the Self-Service Coffee Kiosk System. The kiosk has a small and clear workflow, so separating the system into presentation, business, and data layers makes the design easy to understand and manage.

Reason 1 - Clear separation of responsibilities

The touch-screen interface only handles customer interaction, while the Order Manager controls the ordering flow. Payment, receipt printing, and menu/pricing storage are handled by separate components. This keeps each part focused on one job and makes changes easier. For example, the user interface can be improved without changing the database logic.

Reason 2 - Suitable for the size of the system

This kiosk has only three coffee types, two sizes, one payment mode, and one receipt printer. A microservices design would add extra network communication and deployment complexity, while a client-server design would make the kiosk more dependent on a central server. Layered architecture gives enough structure without making the solution unnecessarily complex.

Security Advantage

The customer-facing interface does not directly access the menu database or payment internals. Requests first go through the business layer, where the Order Manager can validate the order and send payment only to the Payment Service. This reduces direct access to sensitive operations and keeps responsibilities controlled.

Performance Benefit

The kiosk can keep the main components close together and use simple local calls and database queries. This avoids the extra network overhead of a distributed microservices solution, helping menu loading, order processing, payment response, and receipt printing remain fast for customers.

Conclusion

For this scenario, Layered Architecture is a practical choice because it is simple, organized, secure by separation, and fast enough for a self-service kiosk. The component diagram shows the required provided/required interfaces and the main interactions between all five components.